

Novel Hardware Comparison for GPU Enhanced All-to-All

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CCS Concepts: • **Computing methodologies** → **Shared memory algorithms**; • **Hardware** → *Emerging tools and methodologies*.

Additional Key Words and Phrases: Alltoall, Collective Communication, GPU, GH200, H100

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1 Introduction

For our project we aimed to perform benchmarking and comparing MPI communication performance across modern high-performance computing architectures to better understand how hardware and data movement strategies impact scalability/efficiency. Our goal was to clearly see which systems offer real advantages for collective communication and where those differences become most impactful.

2 Background

The Alltoall collective is currently one of the main bottlenecks in Deep Learning Model training [3, 4], and the goal of this report is to analyze the runtime of Alltoall communication with variable algorithms and hardware. The Alltoall collective can be sped up via GPU support [2], and a GPU-enhanced Alltoall can be implemented in a few ways: blocking and non-blocking pair-wise exchange, and with GPU-aware or copy-to-CPU MPI communication strategies. It can also be implemented on multiple different types of GPUs, which in our study we used the NVIDIA H100s and NVIDIA's Superchip, the GH200s.

A pairwise exchange Alltoall involves every process sending a message to every other by receiving from their left, and sending to their right, as pictured in Figure 1 and as described in Algorithm 1.

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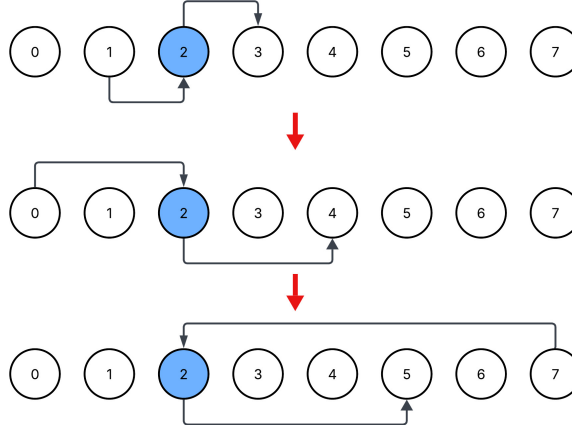


Fig. 1. Pairwise Exchange Visualization

Algorithm 1 Pair Wise Exchange

```

1:  $p \leftarrow$  process rank
2:  $n \leftarrow$  number of processes
3: for  $i \leftarrow 0$  to  $n$  do
4:    $s_{proc} \leftarrow (p + i) \bmod n$ 
5:    $r_{proc} \leftarrow (p + n - i) \bmod n$ 
6:   MPI_Sendrecv()
```

▷ receiving from r_{proc} and sending to s_{proc}

Notice that Algorithm 1 is a blocking implementation of the Alltoall collective, but it can be trivially converted to a non-blocking version by swapping the Sendrecv() call to calls to Isend(), Irecv(), and Waitall(). This is pictured in Algorithm 2.

Algorithm 2 Non-Blocking

```

1:  $p \leftarrow$  process rank
2:  $n \leftarrow$  number of processes
3: for  $i \leftarrow 0$  to  $n$  do
4:    $s_{proc} \leftarrow (p + i) \bmod n$ 
5:    $r_{proc} \leftarrow (p + n - i) \bmod n$ 
6:   MPI_Isend()
7:   MPI_Irecv()
8: MPI_Waitall()
```

▷ sending to s_{proc}
▷ receiving from r_{proc}

Additionally, the distinction between a Copy-to-CPU Alltoall and a GPU-aware Alltoall implementation, is that the copy-to-CPU Alltoall is still reliant on the CPU to perform communication, while a GPU-aware version is transferring data directly from GPU to GPU, and should therefore see noticeable performance increases.

3 Methodologies

Tests were performed on the University of New Mexico's high-performance computing cluster, Easley, to benchmark the performance of NVIDIA's H100 GPU's, and on the University of Illinois Urbana-Champaign's DeltaAI cluster to benchmark the performance of NVIDIA's GH200 super-chips. A pair-wise exchange Alltoall algorithm implemented by Kinkead et al. [1], was used to perform the following micro-benchmark tests with both H100 GPU's and GH200 super-chips:

- (1) GPU-aware pairwise exchange all-to-all
- (2) GPU-aware non-blocking pairwise exchange all-to-all
- (3) Copy-to-CPU pairwise exchange all-to-all
- (4) Copy-to-CPU non-blocking pairwise exchange all-to-all

Note, the H100's were tested with message sizes in the range 1 to 2^{11} double-precision floating point numbers, and the GH200's were tested with message sizes in the range 1 to 2^{17} .

4 Findings

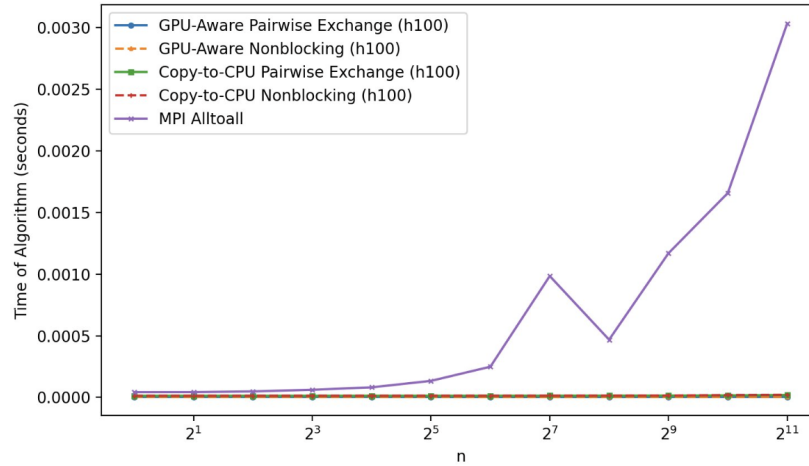


Fig. 2. CPU vs. GPU-Enhanced Alltoall Performance

Figure 2 demonstrates that the pure CPU MPI implementation of Alltoall performs worse than all GPU-enhanced versions.

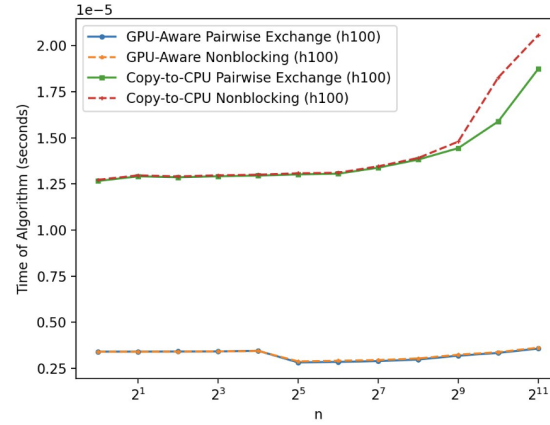


Fig. 3. NVIDIA H100 GPU-Enhanced Alltoall Performance

Figure 3 depicts the performance of the Easley H100 GPUs with Pairwise Exchange and Nonblocking performing quite similarly with the main performance difference coming from GPU-Aware v. Copy-to-CPU communication, where GPU-Aware performed better.

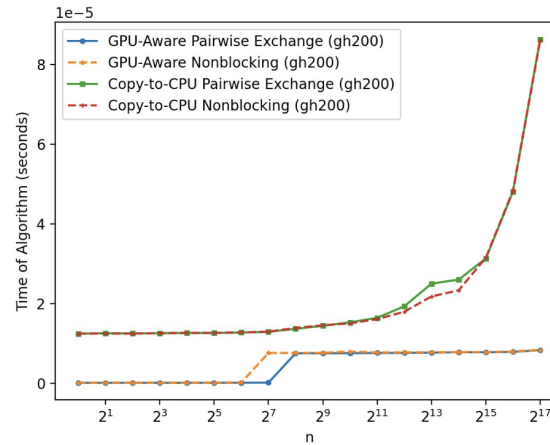


Fig. 4. NVIDIA GH200 GPU-Enhanced Alltoall Performance

The performance on the Delta AI GH200 GPUs with Pairwise Exchange and Nonblocking again performing quite similarly with the main performance difference coming from GPU-Aware v. Copy-to-CPU communication, where GPU-Aware performed better, as pictured in Figure 4.

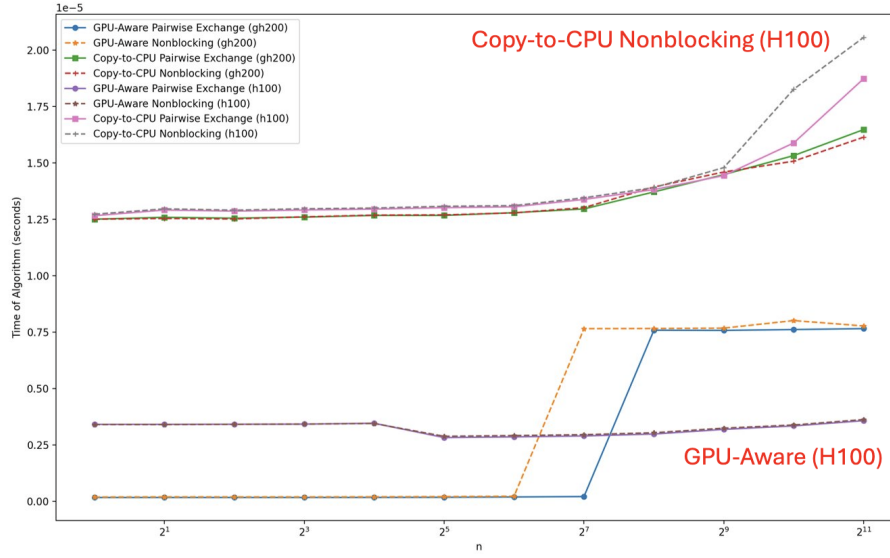


Fig. 5. NVIDIA GH200s v. NVIDIA H100s: Overall View

Figure 5 depicts an overall comparison between GPU-Aware v. Copy-to-CPU on both the NVIDIA H100s and GH200s. It can clearly be seen that GPU-Aware communication performed significantly better than Copy-to-CPU communication. In the comparison between GPUs, it can be seen that for a lower n -size ($< 2^7$) the Delta AI Cluster GH200s outperform the Easley H100s. However, for a larger n -size ($> 2^7$) the H100s outperform the GH200s as the GH200s undergo a hit in performance. We theorize that this could be due to a lower bandwidth capacity available on the GH200 architecture than through the GPU-Direct links for the H100s.

5 Conclusions and Future Work

We tested Alltoall on three different architectures, comparing the performance of each with different implementations. The pure CPU MPI of Alltoall performed the worst (by runtime). The GPU-Aware enhanced Alltoall performed the best, demonstrating that when the hardware for direct GPU-to-GPU communication is available, it will be faster than copying information from a CPU. Variance due to internode communication was not explored in this paper, but should be done in future work. In addition, other types of communication and collectives should be compared.

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